

Results

01 • MIXED ANOVA

between-groups × repeated conditions

Table 1. Descriptives by group × condition

Group	Condition	N	M	SD
control	1	20	59.78	5.21
drug_a	1	20	59.30	6.82
drug_b	1	20	59.58	4.30
control	2	20	61.83	5.06
drug_a	2	20	62.59	6.33
drug_b	2	20	62.84	6.43
control	3	20	63.55	5.51
drug_a	3	20	64.55	6.82
drug_b	3	20	66.39	5.66

Table 2. Mixed ANOVA

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Group (between)	45.68	2	22.84	0.25	.782	0.01 [0.00, 1.00]
Condition (within)	836.30	1.67	500.28	82.56	<.001	0.59 [0.50, 1.00]
Group × Condition	51.09	3.34	15.28	2.52	.056	0.08 [0.00, 1.00]

Table 3. Sphericity (Mauchly's test)

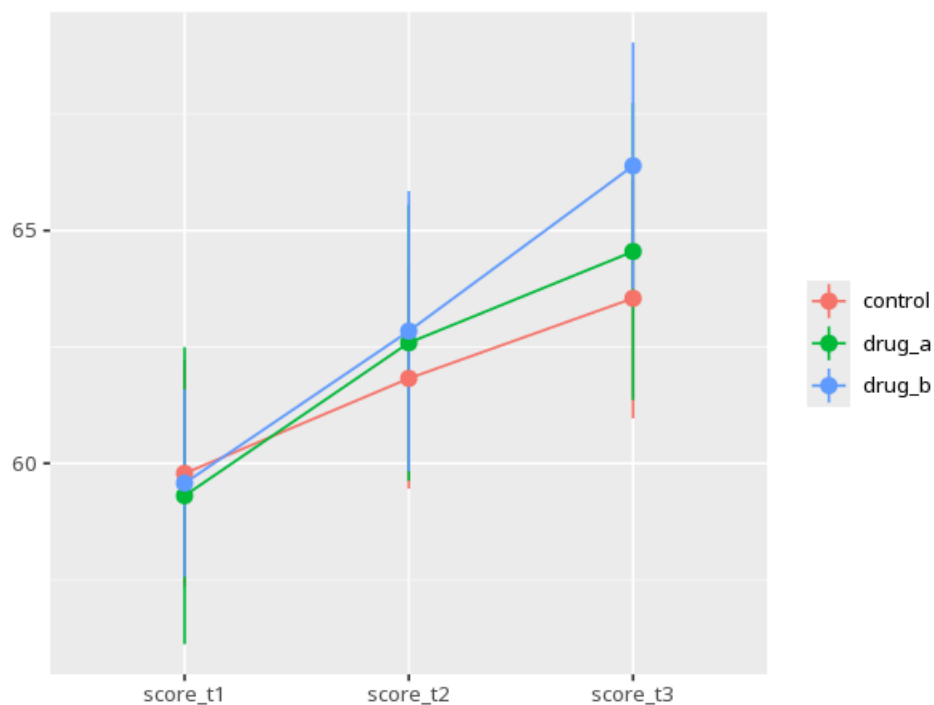
Effect	W	p	GG ϵ	HF ϵ
condition	0.80	.002	0.84	0.86
grp:condition	0.80	.002	0.84	0.86

assumption check: Levene's (equal variances between groups). between and within effects are tested against different error terms; when sphericity is violated the within and interaction F-tests use the Greenhouse–Geisser / Huynh–Feldt correction. Mauchly's test & the GG/HF corrections apply only when the repeated factor has 3+ levels (with 2 levels sphericity is automatically met and this table is omitted). (Levene F=0.60, p=.553)

Table 4. Post-hoc comparisons (condition pairs)

Pair	M _{diff}	SE	P _{adj}	95% CI
score_t1 - score_t2	−2.86	0.42	<.001	[−3.90, −1.83]
score_t1 - score_t3	−5.27	0.32	<.001	[−6.06, −4.49]
score_t2 - score_t3	−2.41	0.48	<.001	[−3.59, −1.23]

Figure. Means across conditions by group



How to read this test

Read the Group $\times$ Condition interaction first — a significant interaction means the groups changed differently across conditions; only then read the main effects. Check sphericity for the within and interaction terms — if violated, read the corrected F/p. When the group $\times$ condition interaction is significant, the overall condition comparisons in the post-hoc table can mislead — read each group's line on the profile plot instead. Your significance threshold (α) is 0.05.

APA template: A mixed ANOVA yielded a Group $\times$ Condition interaction, $F(3.34, 95.29) = 2.52$, $p = .056$, partial $\eta^2 = .08$ [.00, 1.00].

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